

Week 3 Report

Financial Data Import, Transformation & Time Intelligence

Dataset: Shopify Stock Data (SHOP), 21 May 2015 – 16 Mar 2025

Tool: Microsoft Power BI Desktop (Power Query, DAX, Data Model)

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1. Introduction

1.1 Problem Statement

An investor wants to analyze Shopify stock performance over time and understand changes in stock prices and trading activity.

1.2 Objectives

- Import and prepare time-series stock market data in Power Query.
- Generate a dedicated Date Dimension Table (Dim_Date) using DAX (CALENDARAUTO()).
- Set up a 1-to-many active relationship between Dim_Date and the stock price fact table.
- Create core DAX financial measures: Open Price, Close Price, High Price, Low Price, Trading Volume and Daily Return %.

1.3 Tasks & Deliverables

- Generate a Date Table marked as an official Date Table in Power BI.
- Write DAX measures using time-intelligence functions (DATEADD, TOTALYTD, SAMEPERIODLASTYEAR).
- Create column hierarchies for Year-Quarter-Month-Day drill-downs.
- Build a structured data model optimized for financial analytics.

1.4 Dataset Description

The source file (SHOP_2015-05-21_2025-03-16) contains daily Shopify trading records with eight columns:

Column	Description
date	Trading date of the record
open	Price at market open
high	Highest price during the trading day
low	Lowest price during the trading day
close	Price at market close
adj_close	Closing price adjusted for splits and dividends
volume	Number of shares traded during the day

2. Data Import & Preparation (Power Query)

The dataset was loaded into Power Query Editor, where the data was profiled and cleaned before loading to the model. Column profiling on the top 1,000 rows showed 100% valid values and 0% errors or empties across all numeric columns, so no rows needed to be removed.

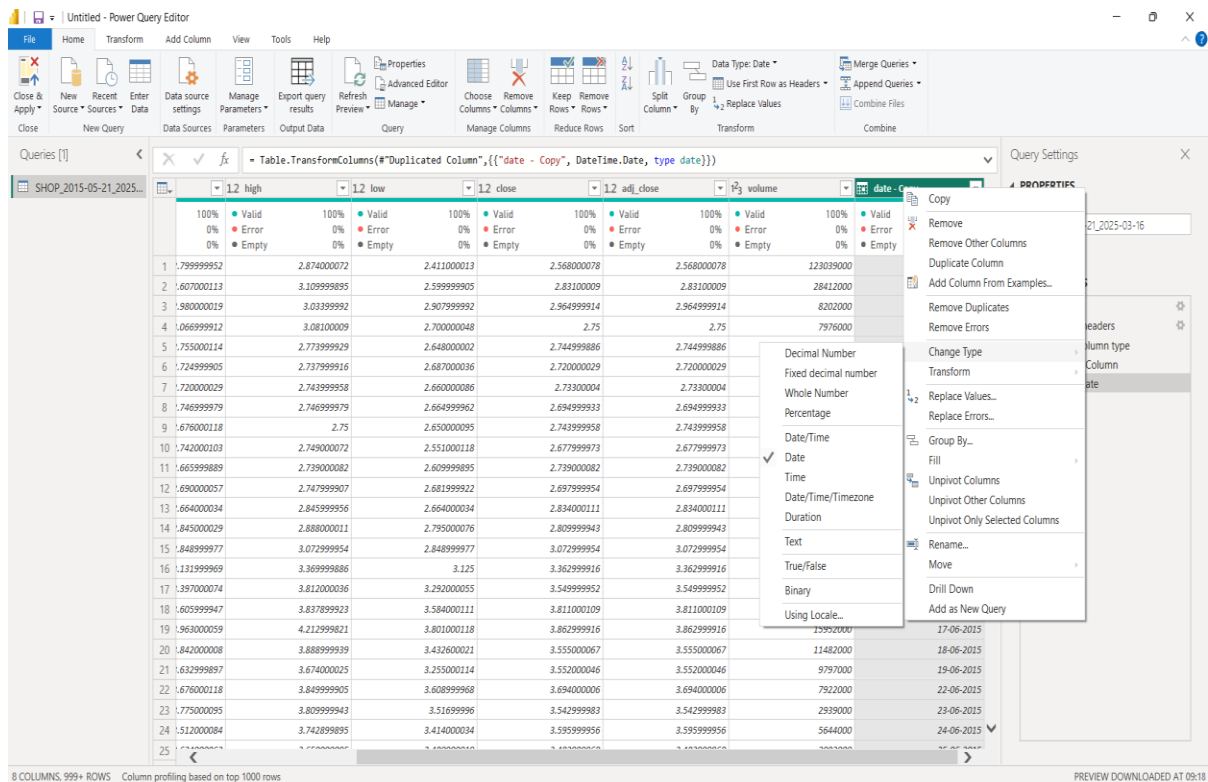
The following Applied Steps were performed:

1. **Source:** connected to the SHOP stock data file.
2. **Promoted headers:** first row used as column headers (date, open, high, low, close, adj_close, volume).
3. **Changed column type:** price columns set to Decimal Number and volume to Whole Number.
4. **Duplicated Column:** the date column was duplicated as date - Copy.
5. **Extracted Date:** the copy was converted from Date/Time to Date type (DateTime.Date), removing the time portion so it can be matched cleanly against the Date Table.

Table: TransformColumns(*Duplicated Column*, [{"date - Copy", DateTime.Date, type date}])

	1.2 high	1.2 low	1.2 close	1.2 adj_close	1.2 volume	date - Copy
1	1.79999952	2.87400072	2.41100013	2.56800078	2.56800078	123039000
2	1.607000113	3.109999895	2.599999905	2.83100009	2.83100009	28412000
3	1.980000019	3.03399992	2.907999992	2.964999914	2.964999914	8202000
4	1.066999912	3.08100009	2.700000048	2.75	2.75	7976000
5	1.755000114	2.773999929	2.648000002	2.744999886	2.744999886	4000000
6	1.724999905	2.737999916	2.687000036	2.720000029	2.720000029	2554000
7	1.720000029	2.743999958	2.660000086	2.733000004	2.733000004	2569000
8	1.746999979	2.746999979	2.664999962	2.694999933	2.694999933	3798000
9	1.676000118	2.75	2.650000095	2.743999958	2.743999958	3024000
10	1.742000103	2.749000072	2.551000118	2.677999973	2.677999973	3226000
11	1.665999889	2.739000082	2.609999895	2.739000082	2.739000082	1294000
12	1.690000057	2.747999907	2.681999922	2.697999954	2.697999954	2809000
13	1.664000034	2.845999956	2.664000034	2.834000111	2.834000111	3375000
14	1.845000029	2.888000011	2.795000076	2.809999943	2.809999943	2734000
15	1.848999977	3.072999954	2.848999977	3.072999954	3.072999954	4606000
16	1.312999969	3.369999886	3.125	3.362999916	3.362999916	8474000
17	1.397000074	3.812000036	3.292000055	3.549999952	3.549999952	16943000
18	1.605999947	3.837899923	3.584000111	3.811000109	3.811000109	11095000
19	1.963000059	4.212999821	3.801000118	3.862999916	3.862999916	15952000
20	1.842000008	3.888999939	3.432600021	3.555000067	3.555000067	11482000
21	1.632999897	3.674000025	3.255000114	3.552000046	3.552000046	9797000
22	1.676000118	3.849999905	3.608999968	3.694000006	3.694000006	7922000
23	1.775000095	3.809999943	3.516999996	3.542999983	3.542999983	2939000
24	1.512000084	3.742899895	3.414000034	3.595999956	3.595999956	5644000
25						

The data type of date - Copy was changed to Date using Change Type → Date, as shown below. Using a pure Date type (without time) is essential for a valid relationship with the date dimension.



3. Date Dimension Table (Dim_Date)

A dedicated Date Table was created in the Modeling tab (Modeling → New table). The table is generated with the DAX function CALENDAR(), using the earliest and latest dates in the stock data as its start and end points. This produces one continuous row for every calendar date in the dataset (21 May 2015 to 16 Mar 2025) with no gaps, which is required for time-intelligence functions to work correctly. (CALENDARAUTO() can also be used and gives the same continuous-date result, extended to whole calendar years.)

Step 1: Create the Date Table

```
Date Table =
CALENDAR (
    MIN ( 'SHOP_2015-05-21_2025-03-16'[date] ),
    MAX ( 'SHOP_2015-05-21_2025-03-16'[date] )
)
```

The following calculated columns were then added to the Date Table (Date Table → New column):

Step 2: Add Year

```
Year = YEAR ( 'Date Table'[Date] )
```

Step 3: Add Quarter

```
Quarter = "Q" & QUARTER ( 'Date Table'[Date] )
```

Returns Q1, Q2, Q3 and Q4.

Step 4: Add Month

```
Month = FORMAT ( 'Date Table'[Date], "MMMM" )
```

Returns the month name: January, February, March ... December.

Step 5: Add Month Number

```
Month Number = MONTH ( 'Date Table'[Date] )
```

Returns 1 to 12, and is used to sort Month chronologically.

Step 6: Add Day

```
Day = DAY ( 'Date Table'[Date] )
```

Returns the day of the month, 1 to 31.

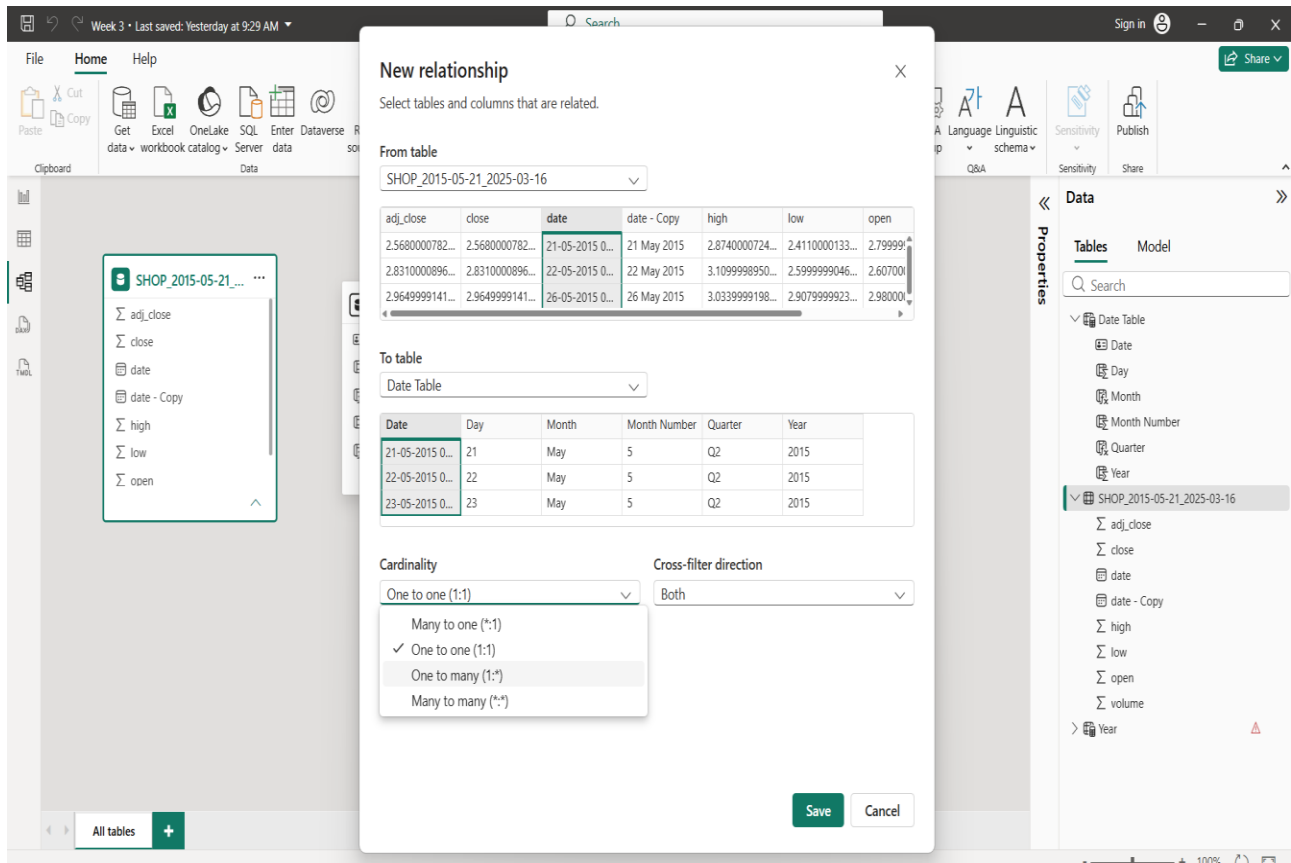
Step 7: Mark as Date Table

The table was marked as the official date table (Table tools → Mark as date table) using the Date column, so that Power BI recognizes it for time-intelligence calculations. Month was also sorted by Month Number so months appear in calendar order rather than alphabetically.

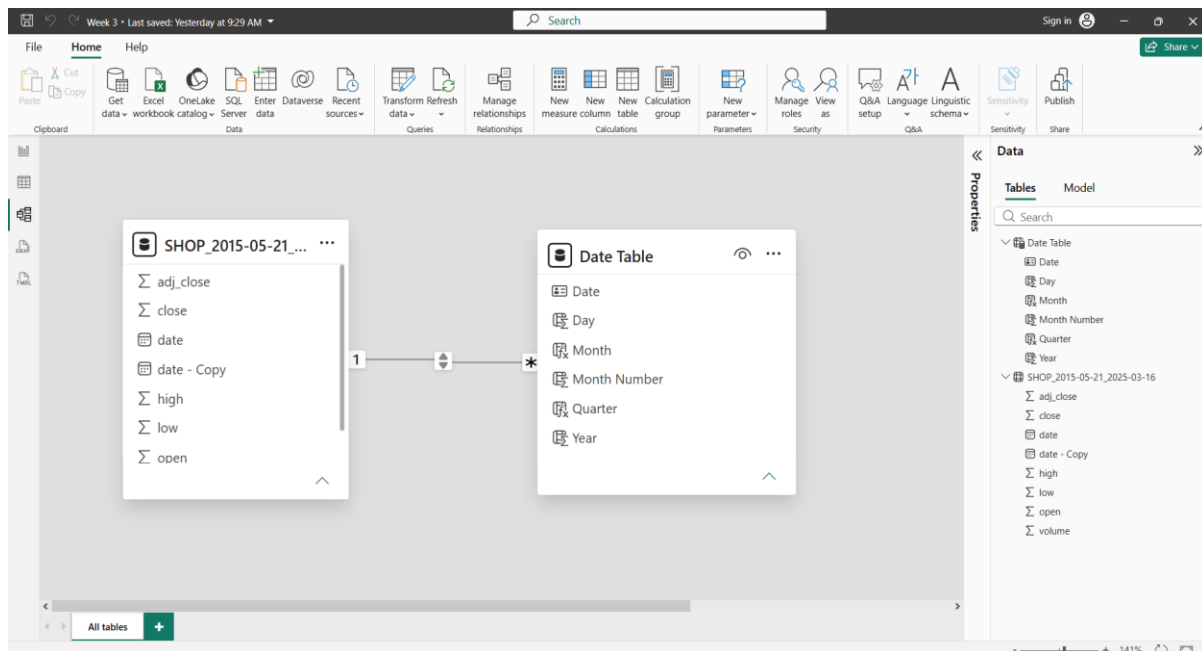
Column	Purpose
Date	Primary date key (unique, continuous)
Year	Calendar year, e.g. 2015
Quarter	Quarter label, e.g. Q2
Month	Month name, e.g. May
Month Number	Numeric month, used to sort Month chronologically
Day	Day of the month

4. Data Model & Relationship

A relationship was created between the fact table (SHOP_2015-05-21_2025-03-16) and the Date Table using the date column from the fact table and the Date column from the Date Table. The New relationship dialog is shown below.



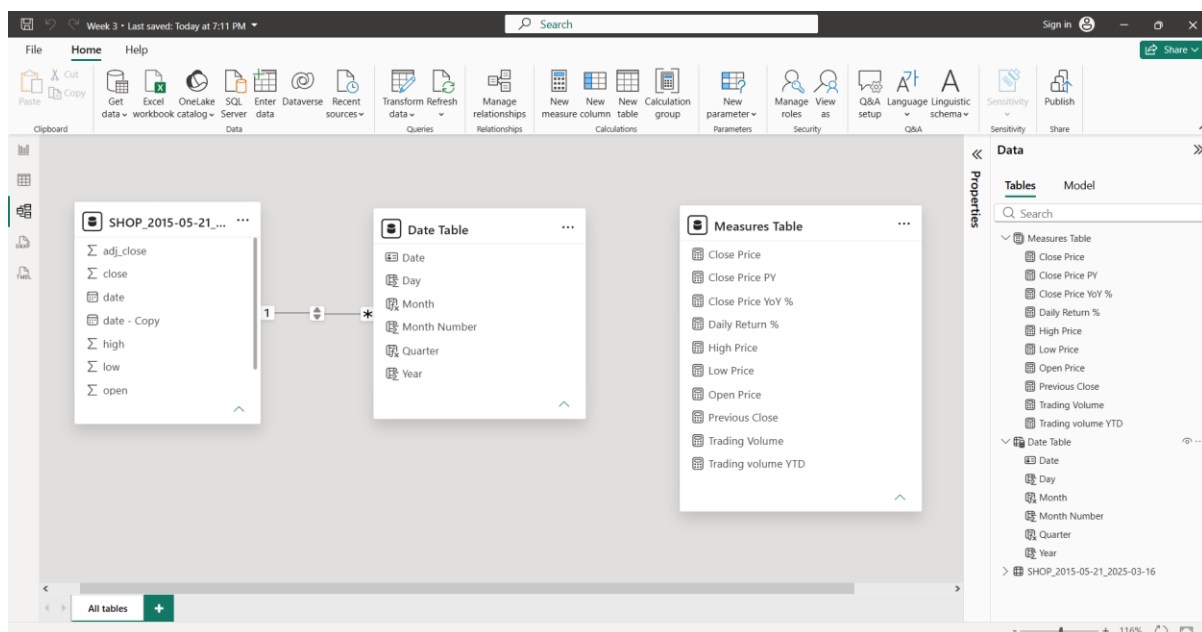
The relationship was configured as a one-to-many (1:*) active relationship: one row per date in the Date Table relates to many rows in the stock table, with the Date Table on the "one" side. The cross-filter direction was kept as single so that the Date Table filters the stock data. The resulting model is shown below.



Why it matters: a proper star-schema style relationship allows every visual and time-intelligence measure to be filtered consistently by Year, Quarter, Month and Day.

5. DAX Measures

A separate Measures Table was created to keep all calculations organized and separate from the raw data. It contains core financial measures and time-intelligence measures.



5.1 Core Financial Measures

```
Open Price = AVERAGE ( 'SHOP_2015-05-21_2025-03-16'[open] )

Close Price = AVERAGE ( 'SHOP_2015-05-21_2025-03-16'[close] )

High Price = MAX ( 'SHOP_2015-05-21_2025-03-16'[high] )

Low Price = MIN ( 'SHOP_2015-05-21_2025-03-16'[low] )

Trading Volume = SUM ( 'SHOP_2015-05-21_2025-03-16'[volume] )
```

Measure	Meaning
Open Price	Average opening price for the selected period
Close Price	Average closing price for the selected period
High Price	Highest price reached in the selected period
Low Price	Lowest price reached in the selected period
Trading Volume	Total number of shares traded in the selected period

5.2 Time-Intelligence Measures

These use DATEADD, SAMEPERIODLASTYEAR and TOTALYTD, which rely on the marked Date Table. DIVIDE() is used with an alternate result of 0 so that a zero or blank denominator does not cause an error.

```
Previous Close =
CALCULATE (
    [Close Price],
    DATEADD ( 'Date Table'[Date], -1, DAY )
)

Daily Return % =
DIVIDE (
    [Close Price] - [Previous Close],
    [Previous Close],
    0
)

Close Price PY =
CALCULATE (
    [Close Price],
    SAMEPERIODLASTYEAR ( 'Date Table'[Date] )
)
```

```

Close Price YoY % =
DIVIDE (
    [Close Price] - [Close Price PY],
    [Close Price PY],
    0
)

Trading volume YTD =
TOTALYTD ( [Trading Volume], 'Date Table'[Date] )

```

Measure	Function Used	Business Question Answered
Previous Close	DATEADD	What was the closing price on the previous day?
Daily Return %	DATEADD (via Previous Close)	How much did the stock gain or lose day over day?
Close Price PY	SAMEPERIODLASTYEAR	What was the close in the same period last year?
Close Price YoY %	SAMEPERIODLASTYEAR (via Close Price PY)	How has the price changed compared with last year?
Trading volume YTD	TOTALYTD	How many shares have traded so far this year?

6. Hierarchies

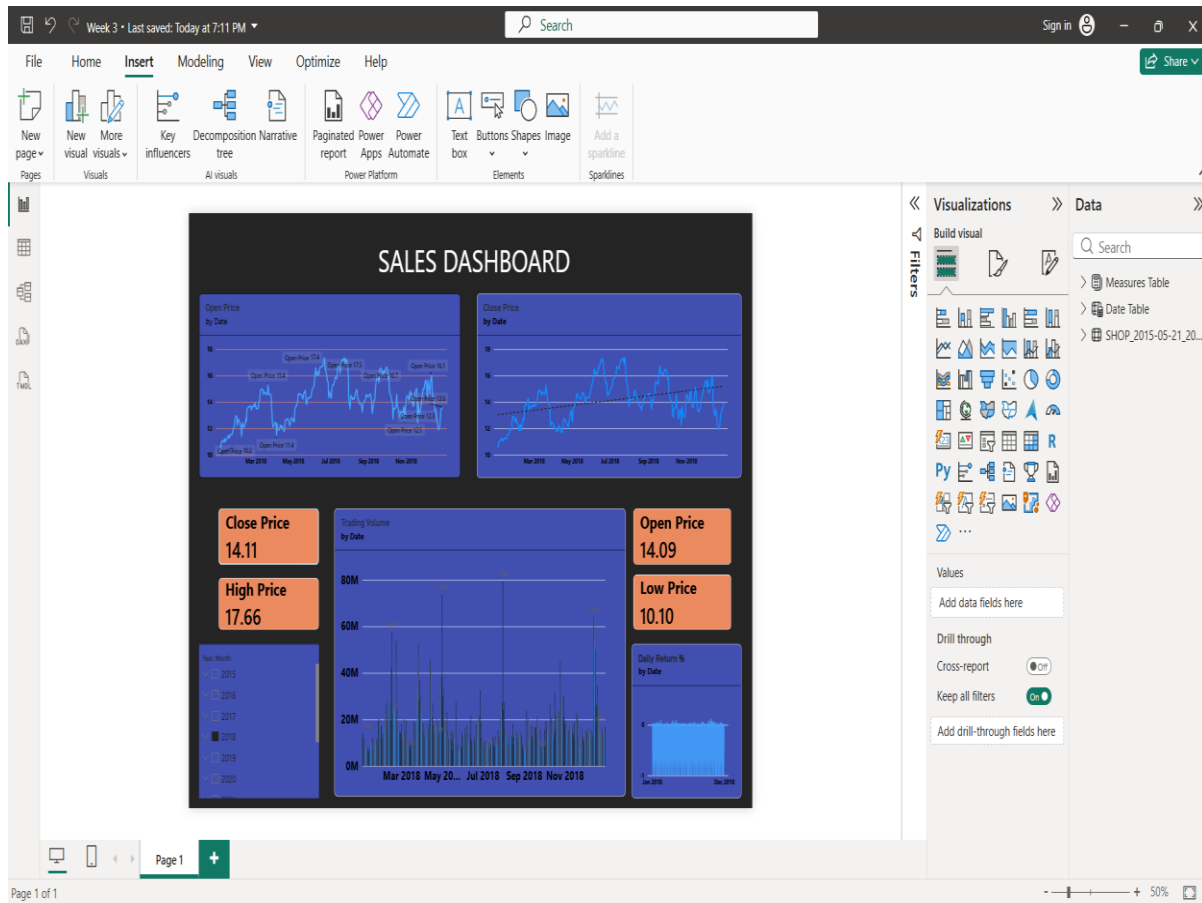
A Year-Quarter-Month-Day hierarchy was built on the Date Table (Year → Quarter → Month → Day). This enables drill-down in visuals, so an investor can start with an annual view and progressively drill into quarters, months and individual days. The Year → Month hierarchy is used in the slicer on the dashboard.

7. Dashboard

Using the model and measures above, an interactive dashboard was built for stock analysis. It includes:

- **KPI cards:** Close Price, Open Price, High Price and Low Price.
- **Open Price by Date:** line chart with data labels.
- **Close Price by Date:** line chart with a trend line.
- **Trading Volume by Date:** column chart showing trading activity and volume spikes.

- **Daily Return % by Date:** area chart showing day-to-day volatility.
- **Year/Month slicer:** based on the date hierarchy, to filter the whole page (2018 is selected in the screenshot).



Key Observations (2018 selection)

- Average close of about 14.11 and average open of about 14.09 show prices closing marginally above opening levels.
- The price ranged from a low of 10.10 to a high of 17.66 during the year, showing considerable price movement.
- Trading volume was mostly steady with several sharp spikes, typically corresponding to periods of large price movement.
- Daily Return % stays in a narrow band around zero for most days, with occasional larger swings.

8. Conclusion

This week covered the full workflow of preparing time-series financial data for analysis. Shopify stock data was cleaned in Power Query, a dedicated Date Table was generated with DAX and marked as the official date table, and a one-to-many relationship was created to form a clean data model. Core financial measures and time-intelligence measures (DATEADD, SAMEPERIODLASTYEAR, TOTALYTD) were written in a dedicated Measures Table, and a Year-Quarter-Month-Day hierarchy enabled drill-downs. The final dashboard lets an investor examine price changes and trading activity over any period, and the model provides a solid base for further financial analytics.